

Data Flow Diagrams & User Stories

Date: 18 June 2026

Project Name: House Rent

Maximum Marks: 4 Marks

DATA FLOW & USER STORIES SPECIFICATION

SYSTEM ARCHITECTURE:

[FRONTEND - React UI]

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HTTP/REST API

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[BACKEND - Express Server]

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MongoDB Queries

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[DATABASE - MongoDB]

PRIMARY DATA FLOWS:

FLOW 1: PROPERTY SEARCH

1. Renter enters filters (location, rent, type)
 2. React sends GET /api/properties?location=Delhi&maxRent=15000
 3. Express receives request, validates JWT
 4. Mongoose queries MongoDB with filter conditions
 5. MongoDB returns matching properties
 6. Express formats JSON response
 7. React updates state and renders property list
- Time: <1 second

FLOW 2: BOOKING REQUEST

1. Renter selects dates on property detail
2. Submits POST /api/bookings with propertyId, dates
3. Express validates tenant role and dates
4. Mongoose creates new Booking document
5. Property status updated to "booked"
6. BookingHistory entry created
7. Response sent to frontend
8. Renter sees success message
9. Booking appears in renter dashboard
10. Owner sees booking request

FLOW 3: OWNER APPROVAL

1. Owner views booking in dashboard
2. Clicks "Approve" button
3. Sends PUT /api/bookings/:id/status
4. Express validates owner role
5. Mongoose updates booking status to "confirmed"
6. Property status confirmed
7. Response to frontend
8. Status updates visible to both tenant and owner

KEY COLLECTIONS:

Users:

Fields: _id, name, email, password, role, approved, location

Indexes: email (unique), role, approved

Properties:

Fields: _id, owner, title, description, location, rentAmount, type, furnishing, amenities, status, createdAt

Indexes: owner, location, status

Bookings:

Fields: _id, property, tenant, startDate, endDate, status, createdAt

Indexes: property, tenant, status

USER STORIES:

USER STORY 1: Tenant Registration

As a renter, I want to register with email and password so I can create account

Acceptance:

Registration form displays

Email validated

Password strength checked

Account created

Redirected to login

Points: 3

USER STORY 2: Search Properties

As a renter, I want to filter properties by location and rent so I find suitable rentals

Acceptance:

Search filters display

Can enter location

Can enter rent range

Results filtered correctly

Matches displayed

Points: 5

USER STORY 3: View Property Details

As a renter, I want to see full property info including owner contact

Acceptance:

Detail page loads

Title, description shown

Owner name and email visible

Amenities listed

Book button available

Points: 3

USER STORY 4: Submit Booking

As a renter, I want to submit booking with dates so owner can review

Acceptance:

Booking form appears

Date pickers work

Submit creates booking

Status set to pending

Owner sees request

Points: 4

USER STORY 5: Owner Approve Booking

As owner, I want to approve bookings so I can confirm rentals

Acceptance:

Owner dashboard shows requests

Can view request details

Can approve/reject

Status updates immediately

Tenant notified

Points: 3

USER STORY 6: Add Property Listing

As owner, I want to list property with details so renters can find it

Acceptance:

Add property form displays

Can enter all details

Can select amenities

Submit creates property
Property appears in search
Points: 4

USER STORY 7: Admin Approve Owner

As admin, I want to approve owners so only legitimate owners can list

Acceptance:

Admin sees pending owners

Can view owner details

Can approve/reject

Approval status updates

Owner can list after approval

Points: 3